Electronics Store Inventory Management System SRS

Project: Electronics Store Inventory Management System

Version: 1.0

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Date: 06-09-2026

Status: Draft - For Review

Revision history

Version	Date	Author(s)	Change Summary	Approval
1.0	05-09-2026	Karan Maheshwari	Introduction + Product Perspective, Major Product Functions + Quality Attributes	Pending
1.0	05-09-2026	Krishitha Kankanala	User Roles, Operating Environment, Constraints + External Interface Requirements + Acceptance Tests	Pending
1.0	05-09-2026	Jayashree D	Components Inventory Management + Add Component + Delete Component + RTM	Pending
1.0	05-09-2026	Hemavathi E	Availability Report + Dashboard + Non-functional Requirements + Security + System Models & Diagrams	Pending

Approvals

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1. Introduction

1.1 Purpose

This document is a Software Requirements Specification (SRS) for the Electronics Store Inventory Management System (EIMS). It defines the functional and non-functional requirements, external interfaces, and verification criteria for a system that helps an electronics store manage components inventory, including adding, deleting, and tracking stock availability through a central dashboard.

1.2 Scope

The system covers management of electronic components inventory (e.g., resistors, capacitors, ICs, sensors, modules, cables, tools, and other store items), including adding new components, deleting/discontinuing components, updating stock quantities, generating availability reports, and a summary dashboard for store owners/managers and staff. It excludes point-of-sale (billing) integration with external payment gateways and supplier-side procurement systems, though interfaces to such systems may be considered for future scope.

1.3 Audience

Developers, QA Engineers, Project Guides/Evaluators, Store Owner/Manager, and Store Staff (end users).

1.4 Definitions

List of acronyms: EIMS (Electronics Store Inventory Management System), SRS, UI, API, DB, SKU (Stock Keeping Unit), CRUD (Create/Read/Update/Delete), RBAC (Role-Based Access Control), JWT (JSON Web Token), ORM (Object-Relational Mapping), REST (Representational State Transfer).

2. Overall description

2.1 Product perspective

EIMS is a standalone, web-based application used within a single electronics retail store. It comprises a front-end user interface, a back-end application server exposing REST APIs, and a relational/NoSQL database that stores component, category, supplier, and transaction data. It may be implemented using Python (Django), JavaScript (MERN stack — MongoDB, Express.js, React, Node.js), or any other suitable technology stack chosen by the development team, provided it exposes a REST API architecture.

2.2 Major product functions (detailed)

- Components Inventory management (view, search, filter, sort, update)
- Add new component to inventory
- Delete / discontinue an existing component
- Availability report generation (in-stock, low-stock, out-of-stock)
- Dashboard with summary widgets and charts
- User authentication and role-based access control
- Audit logging of inventory changes

2.3 User roles and characteristics (expanded)

- Admin / Store Manager: full access — add, edit, delete components, view all reports and manage users.
- Sales / Store Staff: limited access — view inventory, search components, view availability report and dashboard (read-only), update stock quantity on sale.
- Supplier (optional/future): may receive low-stock notifications for reorder coordination.

2.4 Operating environment

Web application accessible via desktop or tablet browsers within the store network or over the internet; deployed on a cloud or on-premise server with a relational/NoSQL database; requires a stable internet or LAN connection and a modern browser (Chrome, Firefox, Edge).

2.5 Constraints

- Must run on commonly available browsers without proprietary plugins.
- Database must maintain referential integrity between components, categories, and suppliers.

- Development limited to the chosen stack's standard libraries/frameworks (Django/DRF or Node/Express/React/MongoDB, or equivalent).
- Must be deliverable and demonstrable within the academic/project timeline.

3. External interface requirements

3.1 User interfaces

Primary UI: responsive web dashboard with a clear navigation menu (Inventory, Add Component, Reports, Dashboard, Users). Forms use inline validation messages; tables support search, filter, and pagination.

3.2 Hardware interfaces

- Barcode scanner (optional) for quick component lookup/entry
- Standard keyboard, mouse, and touchscreen (tablet) input
- Printer for exporting/printing availability reports (optional)

3.3 Software interfaces

- Backend REST API (JSON over HTTPS) for all CRUD operations on components, categories, and users
- Database interface (SQL/NoSQL) via ORM/ODM for persistence
- Optional email/SMS notification service for low-stock alerts
- Optional export service for PDF/Excel/CSV report generation

3.4 Communications

- HTTPS (TLS 1.2+) enforced for all client-server communication
- JWT-based session tokens with expiry for authenticated API calls

4. System features (detailed)

Each requirement below includes acceptance criteria and a reference test case. IDs follow EIMS-F-###.

4.1 Components Inventory Management

Description: View, search, filter, sort, and update electronic components held in store inventory.

Req ID	Requirement (shall...)	Type	Priority	Source/Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
EIMS-F-001	The system shall display a list of all inventory components with attributes (name, SKU, category, quantity, unit price, supplier, reorder level).	Functional	High	Store Manager	AC-EIMS-F-001: Inventory list loads and displays all active components with correct attributes. Test: TC-Inv-01	Foundational requirement; other inventory features (search, filter, sort, update) depend on this.
EIMS-F-002	The system shall allow users to search components by name, SKU, or category.	Functional	High	Store Staff	AC-EIMS-F-002: Search returns matching components within 2 seconds. Test: TC-Inv-02	Depends on EIMS-F-001 (inventory list) being available.
EIMS-F-003	The system shall allow users to filter components by category, supplier, and stock status.	Functional	Medium	Store Manager	AC-EIMS-F-003: Filtered results match selected criteria. Test: TC-Inv-03	Depends on EIMS-F-001; requires category and supplier master data.
EIMS-F-004	The system shall allow sorting of the component list by quantity, price, or date added.	Functional	Medium	Store Staff	AC-EIMS-F-004: List re-orders correctly per selected sort key. Test: TC-Inv-04	Depends on EIMS-F-001.
EIMS-F-005	The system shall allow authorized users to update component details (quantity, price, supplier, reorder level).	Functional	High	Store Manager	AC-EIMS-F-005: Updated values persist and reflect immediately in inventory list. Test: TC-Inv-05	Edit access restricted to Admin/Store Manager role.

4.2 Add Component

Description: Add a new electronic component to the store inventory with validated data.

Req ID	Requirement (shall...)	Type	Priority	Source/Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
EIMS-F-006	The system shall provide a form to add a new component with mandatory fields (name, category, quantity, unit price, supplier).	Functional	High	Store Manager	AC-EIMS-F-006: Component saved and appears in inventory list after submission. Test: TC-Add-01	Requires category and supplier master data to be pre-populated.
EIMS-F-007	The system shall auto-generate a unique component ID/SKU for every new component.	Functional	High	System	AC-EIMS-F-007: Each new component receives a unique, non-reused SKU. Test: TC-Add-02	Auto-generated by backend; no dependency on user input.
EIMS-F-008	The system shall validate mandatory fields and reject submission with a clear	Functional	High	QA	AC-EIMS-F-008: Form blocks submission and highlights invalid fields. Test: TC-Add-03	Client-side and server-side validation both required.

Req ID	Requirement (shall...)	Type	Priority	Source/Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
	error message on missing/invalid data.					
EIMS-F-009	The system shall prevent duplicate component entries with the same SKU.	Functional	Medium	Store Manager	AC-EIMS-F-009: Duplicate SKU submission is rejected with an error message. Test: TC-Add-04	Depends on EIMS-F-007 (unique SKU generation).
EIMS-F-010	The system shall allow optional upload of a component image.	Functional	Low	Store Staff	AC-EIMS-F-010: Uploaded image is stored and displayed with the component. Test: TC-Add-05	Optional feature. Max file size 5MB; requires image storage/hosting.

4.3 Delete Component

Description: Remove or discontinue a component from the active inventory in a controlled manner.

Req ID	Requirement (shall...)	Type	Priority	Source/Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
EIMS-F-011	The system shall allow authorized users to delete a component from inventory.	Functional	High	Store Manager	AC-EIMS-F-011: Deleted component no longer appears in the active inventory list. Test: TC-Del-01	Admin/Manager role only
EIMS-F-012	The system shall display a confirmation prompt before permanently deleting a component.	Functional	High	QA	AC-EIMS-F-012: Deletion proceeds only after explicit user confirmation. Test: TC-Del-02	UI-level confirmation dialog; no backend dependency.
EIMS-F-013	The system shall restrict delete access to Admin/Store Manager roles only.	Functional	High	Security	AC-EIMS-F-013: Staff role attempting delete receives an access-denied response. Test: TC-Del-03	Depends on EIMS-SR-003 (role-based access control).
EIMS-F-014	The system shall record an audit log entry (user, timestamp, component) for every deletion.	Functional	Medium	Compliance	AC-EIMS-F-014: Audit log contains an entry matching each deletion event. Test: TC-Del-04	Depends on audit logging mechanism; required for compliance (see EIMS-SR-006).

4.4 Availability Report

Description: Generate reports showing current stock availability across all components.

Req ID	Requirement (shall...)	Type	Priority	Source/Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
EIMS-F-015	The system shall generate a real-time availability report classifying components as in-stock, low-stock, or out-of-stock.	Functional	High	Store Manager	AC-EIMS-F-015: Report reflects current inventory counts against reorder thresholds. Test: TC-Rep-01	Depends on EIMS-F-001 (inventory data) and reorder level field.
EIMS-F-016	The system shall allow the availability report to be filtered by category, supplier, or date range.	Functional	Medium	Store Staff	AC-EIMS-F-016: Filtered report shows only matching components. Test: TC-Rep-02	Depends on EIMS-F-015.

Req ID	Requirement (shall...)	Type	Priority	Source/Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
EIMS-F-017	The system shall allow export of the availability report as PDF, Excel, or CSV.	Functional	Medium	Store Manager	AC-EIMS-F-017: Exported file contains report data matching the on-screen view. Test: TC-Rep-03	Requires export library for PDF/Excel/CSV generation.
EIMS-F-018	The system shall visually highlight components at or below their reorder threshold.	Functional	Medium	Store Manager	AC-EIMS-F-018: Low-stock rows are visually flagged (e.g., color/badge). Test: TC-Rep-04	Depends on reorder level threshold set per component (EIMS-F-005/006).

4.5 Dashboard

Description: Provide a summary view of key inventory metrics for quick decision-making.

Req ID	Requirement (shall...)	Type	Priority	Source/Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
EIMS-F-019	The system shall display summary widgets on the dashboard (total components, total stock value, low-stock count, recently added items).	Functional	High	Store Manager	AC-EIMS-F-019: Widgets show values consistent with underlying inventory data. Test: TC-Dash-01	Depends on EIMS-F-001 and EIMS-F-015 for aggregated data.
EIMS-F-020	The system shall display graphical charts on the dashboard (e.g., stock levels by category, stock trend over time).	Functional	Medium	Store Manager	AC-EIMS-F-020: Charts render correctly and update when inventory changes. Test: TC-Dash-02	Requires a charting library (e.g., Chart.js/Recharts).
EIMS-F-021	The system shall provide role-based dashboard views (full view for Admin, read-only summary for Staff).	Functional	Medium	Security	AC-EIMS-F-021: Staff view excludes edit/delete controls present in Admin view. Test: TC-Dash-03	Depends on EIMS-SR-003 (role-based access control).
EIMS-F-022	The system shall display real-time low-stock alerts/notifications on the dashboard.	Functional	Medium	Store Manager	AC-EIMS-F-022: Alert appears within 5 seconds of stock falling below threshold. Test: TC-Dash-04	Depends on EIMS-F-018 (low-stock detection logic).

5. Non-functional requirements (detailed)

NFRs below are measurable and tied to test plans. IDs EIMS-NF-###.

Req ID	Requirement	Category	Priority	Acceptance criteria / Measurement
EIMS-NF-001	Dashboard and inventory list pages shall load within 3 seconds for 90% of requests under normal load.	Performance	High	90th percentile load time ≤ 3s in production-like test. Test: TC-Perf-01
EIMS-NF-002	The system shall provide 99.5% availability monthly, excluding scheduled maintenance windows.	Reliability	High	Uptime reports show ≥99.5% per month. Test: Ops reports
EIMS-NF-003	The system UI shall be intuitive enough for a new staff member to add or search a component within 5 minutes of first use without formal training.	Usability	Medium	Usability test with new users achieves task completion ≤5 min. Test: TC-UX-01

Req ID	Requirement	Category	Priority	Acceptance criteria / Measurement
EIMS-NF-004	The system shall support at least 10,000 components and 20 concurrent users without performance degradation.	Scalability	Medium	Load test with 10,000 records and 20 concurrent sessions passes. Test: TC-Scale-01
EIMS-NF-005	The system codebase shall be modular (separate layers for UI, API, and data access) and documented to support maintenance.	Maintainability	Medium	Code review confirms layered architecture and API documentation exists. Test: Code review checklist
EIMS-NF-006	The system shall be responsive and usable on both desktop and tablet screen sizes.	Portability/Usability	Low	UI verified on desktop (≥1280px) and tablet (≥768px) viewports. Test: TC-UX-02

5.1 Security

5.1.1 Security Objectives

- Protect the confidentiality and integrity of inventory and user account data against unauthorized access or modification.
- Ensure only authenticated and authorized personnel (by role) can add, edit, or delete inventory components.

5.1.2 Security Requirements

Req ID	Requirement (shall...)	Type	Priority	Acceptance criteria / Test case ref
EIMS-SR-001	The system shall require user authentication (username/password) before granting access to any inventory function.	Security	High	AC-EIMS-SR-001: Unauthenticated requests to protected endpoints are rejected (401). Test: TC-Sec-01
EIMS-SR-002	The system shall store user passwords using a salted one-way hash (e.g., bcrypt/argon2), never in plaintext.	Security	High	AC-EIMS-SR-002: Password field in database is a hash, not plaintext. Test: TC-Sec-02
EIMS-SR-003	The system shall enforce role-based access control (RBAC) distinguishing Admin/Manager and Staff permissions.	Security	High	AC-EIMS-SR-003: Staff-role calls to delete/admin endpoints return 403 Forbidden. Test: TC-Sec-03
EIMS-SR-004	The system shall enforce TLS 1.2+ (HTTPS) for all client-server communication.	Security	High	AC-EIMS-SR-004: Plain HTTP requests are redirected/rejected; TLS handshake verified. Test: TC-Sec-04
EIMS-SR-005	The system shall automatically terminate an inactive user session after 15 minutes.	Security	Medium	AC-EIMS-SR-005: Session token expires and requires re-login after 15 minutes of inactivity. Test: TC-Sec-05
EIMS-SR-006	The system shall validate and sanitize all user input to prevent SQL/NoSQL injection and cross-site scripting (XSS).	Security	High	AC-EIMS-SR-006: Automated security scan (e.g., OWASP ZAP) reports no critical injection/XSS findings. Test: TC-Sec-06

6. Quality attributes & Acceptance tests

- Exit criteria for acceptance: All high-priority functional requirements implemented and verified, no critical NFR failures, and RTM shows all test cases passed.
- Acceptance test suites: Inventory Management, Add Component, Delete Component, Availability Report, Dashboard, Security, and Usability tests.

7. System models and diagrams

7.1 UML Use-Case Diagram — Admin / Store Manager

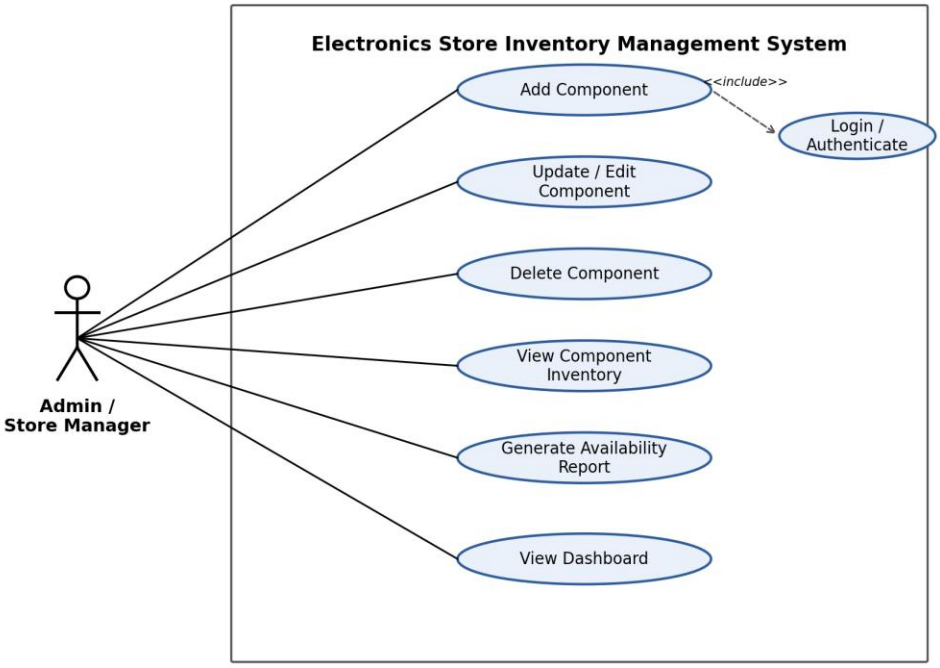


Figure 1: Use-case diagram for Admin / Store Manager role

7.2 UML Use-Case Diagram — Sales / Store Staff

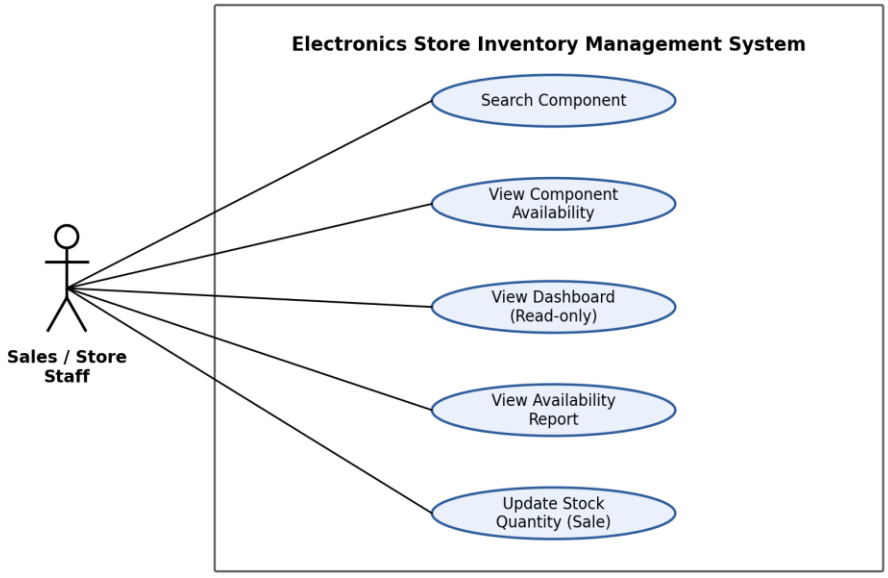


Figure 2: Use-case diagram for Sales / Store Staff role

8. Requirements Traceability Matrix (RTM)

Req ID	Requirement short	Section ref / Design Spec	Module	Test case(s)	Status (N/P/A)	Comments
EIMS-F-001	View inventory list	4.1 / DS-Inv-01	InventoryModule	TC-Inv-01	N	Core inventory listing; verify with sample dataset of 50+ components.
EIMS-F-006	Add new component	4.2 / DS-Add-01	InventoryModule	TC-Add-01	N	Verify SKU uniqueness and mandatory-field validation together.
EIMS-F-011	Delete component	4.3 / DS-Del-01	InventoryModule	TC-Del-01	N	Verify role restriction (EIMS-F-013) alongside deletion.
EIMS-F-015	Generate availability report	4.4 / DS-Rep-01	ReportModule	TC-Rep-01	N	Verify report classification thresholds match reorder levels.
EIMS-F-019	Dashboard summary widgets	4.5 / DS-Dash-01	DashboardModule	TC-Dash-01	N	Verify widget values reconcile with inventory and report data.
EIMS-NF-001	Page load performance	5 / DS-Perf-01	WebUI / API	TC-Perf-01	N	Measure under simulated load with 20 concurrent users.
EIMS-SR-003	Role-based access control	5.1.2 / DS-Sec-01	AuthModule	TC-Sec-03	N	Verify with both Admin and Staff role test accounts.